



Exercise Sheet 02

Published: May 12, 2025

Due: May 19, 2025

Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a jupyter notebook.

1. Draw the diagrams for the following problems:

- (a) Construct a DFA that accepts strings starting with aba , for the language of all the words over $\Sigma = \{a, b\}$. 1P
- (b) Construct a DFA that accepts strings beginning and ending with the same letter, for the language of all the words over $\Sigma = \{a, b\}$. 1P
- (c) Using the automata package presented in the notebooks from the class, provide the code and test your solutions for (a) and (b). 1P

2. Draw the diagrams for the following problems:

- (a) Construct a PDA for the following language $L = \{a^n b^m c^n \mid n \geq 1, m \geq 1\}$ 1P
- (b) Construct a PDA for the following language $L = \{a^n b^{2n} \mid n \geq 1\}$ 1P
- (c) Using the automata package presented in the notebooks from the class, provide the code and test your solutions for (a) and (b). 1P

3. Pumping lemma:

- (a) For any regular language L , there exists a pumping length p such that for any string $s \in L$ with $|s| \geq p$, we can write s as $s = xyz$ where:
 - $|y| \geq 1$
 - $|xy| \leq p$
 - for all $i \geq 0$, the string $xy^i z \in L$

Prove that $L = \{0^n 1^n \mid n \geq 0\}$ is not regular using the Pumping Lemma.

4. Prove that:

- (a) If $M = (Q, \Sigma, \delta, q_0, F)$ is an NFA which accepts the language $L(M)$. There should be equivalent DFA denoted by $M' = (Q', \Sigma', \delta', q'_0, F')$ such that $L(M) = L(M')$. 2P